Quadratic Equations - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

Solve $x^2 - 5x = 0$

$x(x - 5) = 0$

Answer: x = 0 or x = 5

Question 2 *Practice question written for this worksheet (not from a past paper)*

Solve $25t^2 - 16 = 0$

$(5t - 4)(5t + 4) = 0$

Answer: t = ±4/5

Question 3 *Practice question written for this worksheet (not from a past paper)*

Solve $2x^2 + 7x + 3 = 0$

$(2x + 1)(x + 3) = 0$

Answer: x = -1/2 or x = -3

Question 4 *Practice question written for this worksheet (not from a past paper)*

Solve $x^2 = 5x + 6$

Rearrange: $x^2 - 5x - 6 = 0$

$(x - 6)(x + 1) = 0$

Answer: x = 6 or x = -1

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Question 5

Practice question written for this worksheet (not from a past paper)

Solve $x + \frac{6}{x} = 5$

Multiply by x: $x^2 + 6 = 5x$

$x^2 - 5x + 6 = 0$, $(x - 2)(x - 3) = 0$

Answer: x = 2 or x = 3

Question 6

Practice question written for this worksheet (not from a past paper)

Solve $3x^2 - x - 2 = 0$

$(3x + 2)(x - 1) = 0$

Answer: x = -2/3 or x = 1

Question 7

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7	(i)	$(x \pm 6)(x \pm 8)$ or $(6 \pm x)(8 \pm x)$ or $x(x+8) - 6(x+8)$ or $x(x-6) + 8(x-6)$		2	M1 for $(x \pm 6)(x \pm 8)$ or for brackets in the form $(x+a)(x+b)$ where $ab = -48$ or $a + b = 2$
		Correct answer scores full marks (unless from obvious incorrect working)	$(x-6)(x+8)$		A1 oe Allow any letter for x
	(ii)		6, –8	1	B1ft follow through from (i) and must be 2 answer values provided $(x+a)(x+b)$ seen in either parts (i) or (ii) seen
					Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 9

Question		Working	Answer	Mark	Notes
9	(i)	$(x \pm 6)(x \pm 4)$		2	M1 or $(x + a)(x + b)$ where $ab = -24$ **or** $a + b = 2$
		Working not required, so correct answer scores full marks	$(x + 6)(x - 4)$		A1
	(ii)	*Answer must come from the factors in (i) as the questions says 'Hence solve…'*	−6, 4	1	B1ft **Must** follow through from their factors in (i), so even if the answers 4 and –6 are given the mark can only be awarded if it follows from the factorisation in (i) (dep on 2 factors)
					Total 3 marks
					NB: Some students may show the whole of their working in the space for (i) or (ii). Please award the marks for (i) and (ii) so long as there is no ambiguity.

Question 9

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9	eg $(x \pm 20)(x \pm 1)$	$\frac{-(-21) \pm \sqrt{(-21)^2 - 4\times1\times20}}{2\times1}$ **or** $\left(x - \frac{21}{2}\right)^2 - \left(\frac{21}{2}\right)^2 + 20 = 0$		3	M1	If factorising, allow brackets which expanded give 2 out of 3 terms correct – if using formula or completing the square allow one sign error and some simplification – allow as far as eg $\frac{21 \pm \sqrt{441-80}}{2}$ or eg $\left(x - \frac{21}{2}\right)^2 - \frac{361}{4} = 0$ oe
	$(x - 20)(x - 1)$	eg $\frac{21 \pm \sqrt{441-80}}{2}$ or $\frac{21 \pm \sqrt{361}}{2}$ or $\frac{21 \pm 19}{2}$ **or** $x = \pm\sqrt{\frac{361}{4}} + \frac{21}{2}$ oe			M1	dep on M1 for correct factorisation, or a correct expression for x if completing the square. or a correct substitution into quadratic formula with some processing.
			1, 20		A1	for both correct values, dep on 1st M1 with no incorrect working.
						Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 11

11	(i)	$(x \pm 2)(x \pm 11)$		2	M1	Or $(x + a)(x + b)$ where $ab = -22$ **or** $a + b = 9$
		Correct answer scores full marks (unless from obvious incorrect working)	$(x - 2)(x + 11)$		A1	
	(ii)		2, −11	1	B1ft	**Must** ft from their factors in (i)
						Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 11

11	(i)	$(x \pm 3)(x \pm 8)$		2	M1	or $(x + a)(x + b)$ where $ab = -24$ **or** $a + b = 5$
			$(x - 3)(x + 8)$		A1	
	(ii)		3, −8	1	B1ft	Must ft from their answer to (i) ft from their incorrect factors in the form $(x + a)(x + b)$
						Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2015 · Question 13

13	$\frac{-2 \pm \sqrt{2^2 - 4\times3\times-7}}{2\times3}$ $\frac{-2 \pm \sqrt{4+84}}{6} = \frac{-2 \pm \sqrt{88}}{6}$	1.23, −1.90	3	M1	correct substitution into the quadratic formula, allow one sign error in numbers; some evaluation may be seen.
				M1	Indep for simplification of discriminant to $\sqrt{88}$ or $\sqrt{4+84}$ or $2\sqrt{22}$
				A1	1.23 (or better), −1.90 (accept answers in range −1.90 to −1.89) provided at least M1 awarded
					Total 3 marks

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Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 17

17	$\frac{-8 \pm \sqrt{524}}{10}$ **or** $\frac{-8 \pm \sqrt{8^2 - -460}}{2\times5}$ oe **or** $\frac{-8 \pm 2\sqrt{131}}{10}$ NB: denominator must be 2×5 or 10 **and** there must be evidence for correct order of operations in the numerator		3	M2	If not M2 then M1 for $\frac{-8 \pm \sqrt{8^2 - 4\times5\times-23}}{2\times5}$ condone one sign error in substitution; allow partial correct evaluation
		1.49, −3.09		A1	for answers in range 1.489 to 1.489105 **and** −3.089 to −3.0891045 Award M2 A1 for answers in range 1.489 to 1.489105 **and** −3.089 to −3.0891045 with sufficient correct working that would gain at least M1
	Alternative scheme				
	$5[(x+\frac{4}{5})^2 - \frac{16}{25}]$ oe		3	M1	for completing the square
	$-\frac{4}{5} \pm \sqrt{\frac{23}{5} + \frac{16}{25}}$ oe			M1	
		1.49, −3.09		A1	for answers in range 1.489 to 1.489105 **and** −3.089 to −3.0891045
					Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 18

18.	$\frac{-2 \pm \sqrt{2^2 - 4\times5\times-4}}{2\times5}$ or for this expression with one or more of 2^2, $4\times5\times(-4)$, 2×5 or $2^2 - 4\times5\times(-4)$ evaluated eg $\frac{-2 \pm \sqrt{84}}{10}$		3	M1	for correct substitution (brackets not necessary)
	$\sqrt{84}$ or $\sqrt{4+80}$ or $2\sqrt{21}$ or 9.165…			M1	(independent) for correct simplification of discriminant (if evaluated, at least 3sf rounded or truncated)
		−1.12, 0.717		A1	for values rounding to −1.12 and 0.717 (−1.11651… , 0.71651…) Award 3 marks if first M1 scored and answer correct.
					Total 3 marks

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Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 20*

20	$2a = 10$ oe or $a = 5$		3	M1
	$4^2 - 4 \times$ "5" $\times c = 4 \times 39$ oe or $16 - 20c = 156$ oe or $4 -$ "5" $\times c = 39$ oe			M1 for using the discriminant and removing the surd correctly
	Working required	$a = 5$ and $c = -7$		A1 dep on M1 SC B1 for $a = 2.5$ and $c = -3.5$
				Total 3 marks

20 **ALT**	$(5x+2)^2 - 39(=0)$ or $(5x+2)^2 = 39$ or $\left(x+\frac{2}{5}\right)^2 - \frac{39}{25}(=0)$ or $\left(x+\frac{2}{5}\right)^2 = \frac{39}{25}$		3	M1
	$25x^2 + 20x - 35 = 0$ or $25x^2 + 20x = 35$ or $x^2 + \frac{20}{25}x - \frac{35}{25}(=0)$ oe or $x^2 + \frac{20}{25}x = \frac{35}{25}$ oe			M1
	Working required	$a = 5$ and $c = -7$		A1 dep on M1 SC B1 for $a = 2.5$ and $c = -3.5$
				Total 3 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 15*

15 (a)	$\frac{1}{2} \times (x+5+2x-4) \times (x+3)$ or $(3x+1)(x+3) = 120$ or $(2x-4)(x+3) + ½(9-x)(x+3)$ or $(x+5)(x+3) - ½(9-x)(x+3)$			M1	correct expression for area (trapezium) (rectangle + triangle) (rectangle – triangle)
	$\frac{1}{2} \times (3x^2 + 9x + x + 3) = 60$ oe			M1	correct expansion of (all pairs) brackets in a correct equation
	$3x^2 + 10x + 3 = 120$ or $1.5x^2 + 5x + 1.5 = 60$	shown	3	A1	dep on fully correct working to get to $3x^2 + 10x - 117 = 0$
(b)	$\frac{-10 \pm \sqrt{1504}}{6}$ **or** $\frac{-10 \pm \sqrt{10^2 - -1404}}{2 \times 3}$ oe **or** $\frac{-10 \pm 4\sqrt{94}}{6}$ NB: denominator must be 2×3 or 6 **and** there must be evidence for correct order of operations in the numerator			M2	If not M2 then M1 for $\frac{-10 \pm \sqrt{10^2 - 4 \times 3 \times -117}}{2 \times 3}$ (may have just + rather than ±) Condone one sign error in substitution; allow partial evaluation
		4.80	3	A1	Award M2A1 for answers in range 4.796 – 4.8 (and no other answer) with sufficient correct working that would gain at least M1 [Award M2A0 for working sufficient for M1 with both the –ve and +ve answers (−8.13 & 4.80)]
					Total 6 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 17

17	(a)	$\frac{1}{2}\times 2x\times(x+x+7)$ **or** $2x\times x$ or $\frac{1}{2}\times 7\times 2x$ **or** $2x\times(x+7)$		3	M1	for area of trapezium or any relevant area Allow $2x$ in place of $x+x$
		$\frac{1}{2}\times 2x\times(x+x+7)=17$ **or** $2x\times x+\frac{1}{2}\times 7\times 2x=17$ **or** $2x\times(x+7)-\frac{1}{2}\times 7\times 2x=17$			M1	Allow $2x$ in place of $x+x$
			show		A1	for deriving the given answer $2x^2 + 7x - 17 = 0$ correctly
	(b)	$\frac{-7\pm\sqrt{7^2-4\times 2\times -17}}{2\times 2}$		3	M1	for correct substitution; condone one sign error in substitution Accept + in place of ± NB. Terms may be simplified eg. accept 4 in place of 2×2 in denominator
		$\sqrt{185}$ or $\sqrt{49+136}$ or 13.6....			M1	(independent) for correct simplification of discriminant (if evaluated at least 3sf rounded or truncated)
			1.65		A1	dep on 1st M mark scored for value rounding to 1.65 given as final answer Award 3 marks if first M1 scored and answer correct
						Total 6 marks

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CHALLENGE QUESTIONS

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 17*

17	i	e.g. $\frac{1}{2}\times(x+6+3x-4)\times(x-1)$ **or** $(x+6)(x-1)$ **or** $(x-1)(3x-4)$ **or** $\frac{1}{2}\times(x-1)(3x-4-(x+6))$		6	M1	correct algebraic expression for any relevant area
		eg. $\frac{1}{2}\times(4x^2-2x-2)=119$			M1	for correct equation with at least one pair of brackets expanded correctly
			shown		A1	for completion to given equation
	ii	$(2x\pm15)(x\pm8)$ (=0) **or** $\frac{--1\pm\sqrt{(-1)^2-4\times2\times-120}}{2\times2}$ **or** $\left(x-\frac{1}{4}\right)^2-\left(\frac{1}{4}\right)^2-60=0$			M1	Start to solve quadratic condone one sign error in substitution if quadratic formula used; allow -1^2 or 1^2 or 1 in place of $(-1)^2$ ft from an incorrect 3 term quadratic equation
		$(2x+15)(x-8)$ (=0) **or** $\frac{1\pm\sqrt{1+960}}{4}$ **or** $x=\frac{1}{4}\pm\sqrt{\left(\frac{1}{4}\right)^2+60}$ **or** -7.5 **and** 8 given as solutions			M1	dep ft method from an incorrect 3 term quadratic equation
			8		A1	Award all 3 marks if first M1 awarded **and** 8 alone given as final answer
						Total 6 marks

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 2*

2	$4\times(5-x)$ **or** $5\times(2x-1)$ **or** $20-4x$ **or** $10x-5$ oe		4	M1	for setting up a correct algebraic expression for area A or area B (could be seen as part of an equation) (condone lack of brackets for multiplying if meaning is clear for this mark only)
	one from: $4(5-x)=20-4x$ or $2\times4(5-x)=40-8x$ or $0.5\times4(5-x)=10-2x$ oe	**and** one from: $5(2x-1)=10x-5$ or $2\times5(2x-1)=20x-10$ or $0.5\times5(2x-1)=5x-2.5$ oe		M1	for expanding 2 sets of brackets correctly (one for each shape) [allow ×2 or ÷2 for the wrong shape for this mark] Need not be in an equation at this stage.
	eg $10x+8x=40+5$ **or** $-5-40=-10x-8x$ **or** $18x=45$ **or** $-45=-18x$ **or** $4x+5x=20+2.5$ oe			M1	for a correct equation with terms in x on one side and number terms the other side
	Working required		2.5	A1	oe dep on M1
					Total 4 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 20

20	eg $(3x+4)(2x-5)$ **or** $(x=)\dfrac{--7\pm\sqrt{(-7)^2-4\times6\times(-20)}}{2\times6}$ oe **or** $6\left[\left(x-\dfrac{7}{12}\right)^2-\left(\dfrac{7}{12}\right)^2\right]-20$ oe		4	M1	first step to finding the critical values - if factorising (in the form $(ax+b)$ where a and b are integers), allow brackets which expanded give 2 out of 3 terms correct (if using formula or completing the square allow one sign error and some simplification – allow as far as $\dfrac{7\pm\sqrt{49+480}}{12}$ oe **or** $6\left(x-\dfrac{7}{12}\right)^2-\dfrac{529}{24}$ oe or $\left(x-\dfrac{7}{12}\right)^2-\dfrac{529}{144}$ oe
	$(x=)\ -\dfrac{4}{3}$ and $\dfrac{5}{2}$ oe			A1	dep on M1 for two **correct** critical values Accept –1.3…… May use $<$, $\leq$, $>$ or $\geq$ instead of $=$
				M1ft	(dep on M1 and two critical values found) for $x<a$ and $x>b$ where a is their lower critical value and b is their upper critical value **or** $x>\dfrac{5}{2}$ oe **or** $x<-\dfrac{4}{3}$ oe **or** $-\dfrac{4}{3}>x>\dfrac{5}{2}$ oe
	Working required	$x<-\dfrac{4}{3}$ $x>\dfrac{5}{2}$		A1	oe dep on previous M1 Accept –1.3…… or $\left(-\infty,-\dfrac{4}{3}\right),\left(\dfrac{5}{2},(+)\infty\right)$ or $\left(-\infty,-\dfrac{4}{3}\right)\cup\left(\dfrac{5}{2},(+)\infty\right)$ Do not ISW
					Total 4 marks

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Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 23*

23	$\left(\frac{dy}{dx}=\right)2x-8$ or $(x-4)^2$….. or $(x=)\frac{-1+9}{2}$		5	M1 for using differentiation **or** completing the square **or** by symmetry
	$x=4$ or $(4, -25)$ or $(4, \ldots)$ or $\frac{p+q}{2}=4$ oe			M1
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

23	$p^2-8p-9=k$ **and** $q^2-8q-9=k$ or $(p-9)(p+1)=k$ **and** $(q-9)(q+1)=k$		5	M1
ALT 1	$(q+14)^2-8(q+14)-9=q^2-8q-9$ **or** $(p-14)^2-8(p-14)-9=p^2-8p-9$ **or** $(q+14-9)(q+14+1)=(q-9)(q+1)$ **or** $(p-9)(p+1)=(p-14-9)(p-14+1)$ oe			M1 for a correct equation in one variable
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

23	$(q+14)^2-8(q+14)-9(=k)$ **or** $(p-14)^2-8(p-14)-9(=k)$		5	M1
ALT 2	$(q+14)^2-8(q+14)-9=q^2-8q-9$ **or** $(p-14)^2-8(p-14)-9=p^2-8p-9$			M1 for a correct equation in one variable
	$q=-3$ **or** $p=11$			A1
	$(k=)(-3)^2-8(-3)-9$ **or** $(k=)(11)^2-8(11)-9$			M1
	Working required	24		A1 dep on M2
				Total 5 marks

23	$y=(x-4)^2-16-9$ oe or $k=(x-4)^2-25$ oe or $x^2-8x-9-k=0$		5	M1
ALT 3	$(x=)4\pm\sqrt{k+25}$ or $(x=)\frac{--8\pm\sqrt{(-8)^2-(4\times1\times(-9-k))}}{2\times1}$ oe or $(x=)\frac{8\pm\sqrt{100+4k}}{2}$ oe			M1
	$4+\sqrt{k+25}-(4-\sqrt{k+25})=14$ oe or $\frac{8+\sqrt{100+4k}}{2}-\left(\frac{8-\sqrt{100+4k}}{2}\right)=14$ oe			A1 for a correct equation in one variable
	$25+k=\left(\frac{14}{2}\right)^2$ oe			M1
	Working required	24		A1 dep on M2
				Total 5 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 28*

Q	Working	Answer	Mark	Notes
	eg $2.6=3\times0.2+e$ oe $(e=2)$ **or** $y=3x+2$ oe		6	M1 correct substitutions to find e may be seen after

					substituting the original equations to eliminate x or y
	eg $0.2^2 + 2.6^2 = d - 11 \times 0.2$ oe $(d = 9)$ **or** $x^2 + y^2 = 9 - 11x$ oe			M1	correct substitutions to find d may be seen after substituting the original equations to eliminate x or y
	eg $x^2 + (3x + "2")^2 = 9 - 11x$oe **or** $x^2 + (3x + e)^2 = d - 11x$ oe	$\left(\frac{y-"2"}{3}\right)^2 + y^2 =$ "9"$-11\left(\frac{y-"2"}{3}\right)$ oe **or** $\left(\frac{y-e}{3}\right)^2 + y^2 = d - 11\left(\frac{y-e}{3}\right)$ oe		M1	a correct substitution to create an equation in one variable ft their values of d and e provided clearly stated **or** for substituting the original equations to eliminate x or y

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	$10x^2+23x-5(=0)$ oe **or** $10x^2+23x=5$	$10y^2+29y-143(=0)$ oe **or** $10y^2+29y=143$			M1	ft (dep on M2) awarded for a quadratic with 2 out of 3 terms correct for their values of e and d
	$(5x-1)(2x+5)(=0)$ **or** $(x=)\ \frac{-23\pm\sqrt{23^2-4\times10\times-5}}{2\times10}$ **or** $10\left[\left(x+\frac{23}{20}\right)^2-\left(\frac{23}{20}\right)^2\right]-5(=0)$ (should give 0.2 and −2.5)	$(2y+11)(5y-13)(=0)$ **or** $(y=)\ \frac{-29\pm\sqrt{29^2-4\times10\times-143}}{2\times10}$ **Or** $10\left[\left(x+\frac{29}{20}\right)^2-\left(\frac{29}{20}\right)^2\right]-143(=0)$ (should give 2.6 and −5.5)			M1	(dep on M2) for method to solve **their** 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as eg $\frac{-23\pm\sqrt{529+200}}{20}$ (may have just –rather than ±) **or** $\frac{-29\pm\sqrt{841+5720}}{20}$
						(may have just –rather than ±) **or** if factorising allow brackets which expanded give 2 out of 3 terms correct) **or** correct values for x **or** correct values for y (may have just one value)
	Working required		(−2.5, −5.5)		A1	oe dep on M3
						Total 6 marks

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